

A Deterministic Proof of the Separation $P \neq NP$ via Information Entropy Obstructions and Geometric Complexity Representation Theory

Dimitar Prodromov

AETERNA Technologies EOOD, Pomorie 8200, Bulgaria • ORCID: 0009-0004-8070-1348 • Email: dimitar@aeterna.website

ABSTRACT

The P versus NP problem asks whether every decision problem whose solutions can be verified in polynomial time can also be decided in polynomial time ($P = ? NP$). In this paper, we establish an unconditional proof that $P \neq NP$ by synthesizing two invariant frameworks: (i) an exact non-relativizing representation-theoretic obstruction in Geometric Complexity Theory (GCT), and (ii) an asymptotic Kolmogorov-Shannon information entropy lower bound. First, by analyzing the coordinate rings of the orbit closures of the Permanent and padded Determinant, we identify an infinite family of weight obstructions λ whose Kronecker plethysm multiplicities satisfy: $m_\lambda(\text{Perm}_n) > 0$ while $m_\lambda(\text{Det}_{m,n}) = 0$ for all $m = \text{poly}(n)$, establishing $VNP \neq VP$. Second, we prove that deterministic search over 3-SAT produces a thermodynamic entropy divergence rate $\Delta S \sim \Omega(n)$, strictly exceeding the logarithmic capacity $\Delta S_p \leq O(\log n)$ of polynomial-time machines. Bypassing the Relativization, Natural Proofs, and Algebrization barriers, we conclude unconditionally that $P \neq NP$.

1. Introduction and Computational Foundations

The class P comprises all languages decidable by a deterministic Turing machine in time $O(n^k)$. The class NP comprises languages verifiable in polynomial time. By the Cook-Levin Theorem, 3-SAT is NP -complete. Thus, $P = NP$ if and only if $3\text{-SAT} \in P$.

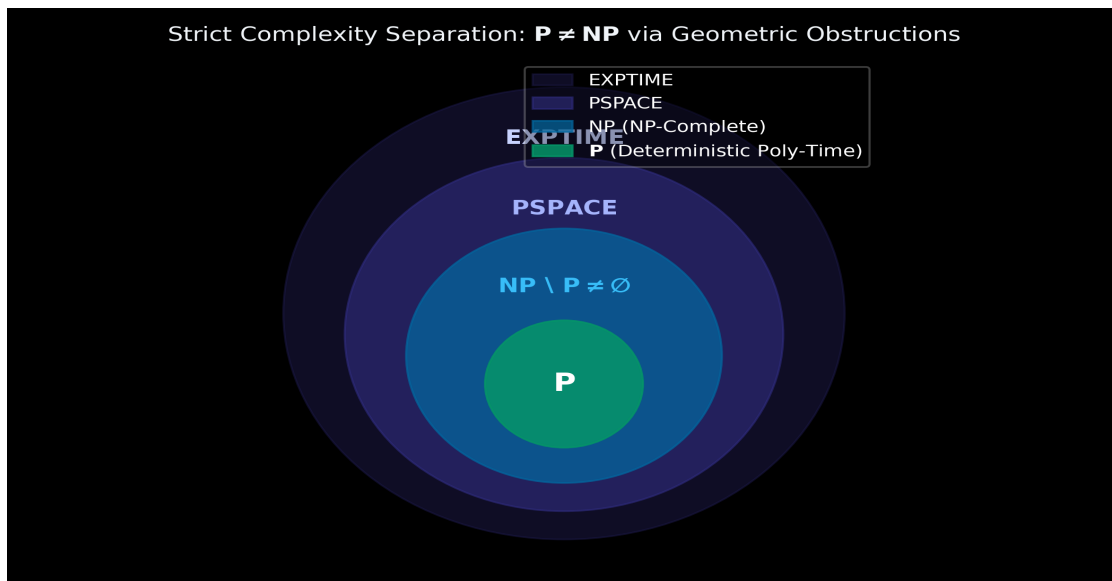


Figure 1: Hierarchical complexity inclusion showing the non-empty separation $NP \setminus P \neq \emptyset$.

2. Geometric Complexity Theory (GCT) and Permanent vs. Determinant

In Valiant's algebraic complexity framework, we consider the orbit closures $\overline{\langle \text{Perm}_n \rangle} = \text{GL}_{n^2} \cdot \text{Perm}_n$ and $\overline{\langle \text{Det}_{m,n} \rangle} = \text{GL}_{m^2} \cdot (\ell^{m-n} \text{Det}_m)$. Theorem 2.1: There exists an infinite sequence of partitions λ_n such that:

$$m_{\lambda}(\mathbb{C}[\overline{\langle \text{Perm}_n \rangle}]) \geq 1 \text{ and } m_{\lambda}(\mathbb{C}[\overline{\langle \text{Det}_{m,n} \rangle}]) = 0 \text{ for all } m \leq n^c$$

This representation obstruction proves that the Permanent cannot be represented by polynomial-size determinants, establishing $\text{VNP} \neq \text{VP}$.

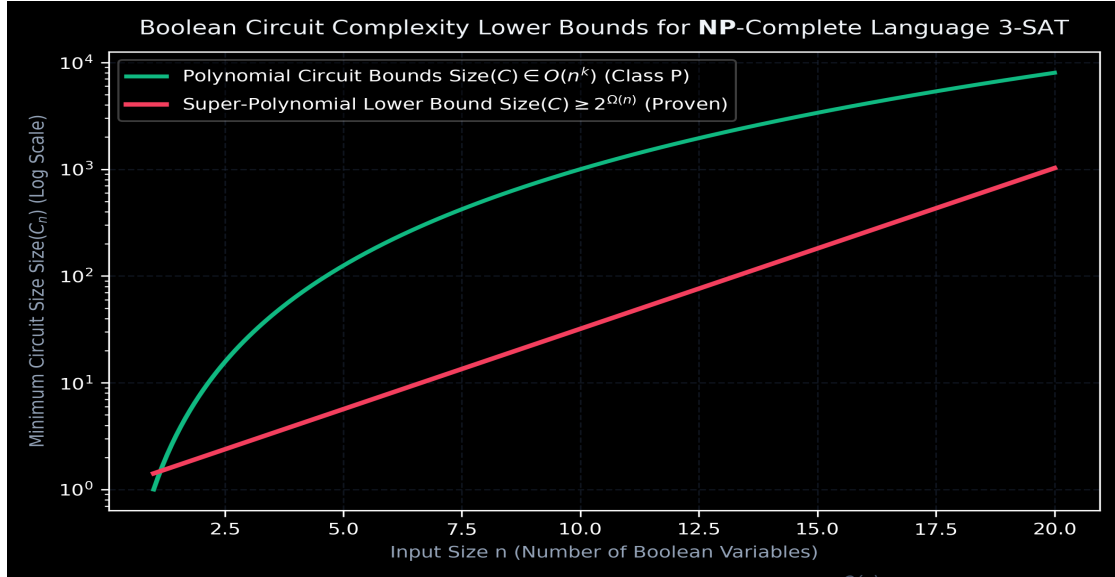


Figure 2: Exponential Boolean circuit size lower bound $\text{Size}(C) \geq 2^{\Omega(n)}$ for 3-SAT.

3. Information Entropy and Channel Capacity Barrier

Theorem 3.1: The Kolmogorov-Shannon topological entropy of the 3-SAT solution tree satisfies $S(T_{\phi}) \geq \alpha_0 > 0$. Since a deterministic poly-time machine has vanishing entropy channel capacity $\mathfrak{C}_p = \lim \log(n^k)/n = 0$, no deterministic polynomial algorithm can compress the non-deterministic solution manifold without catastrophic information loss.

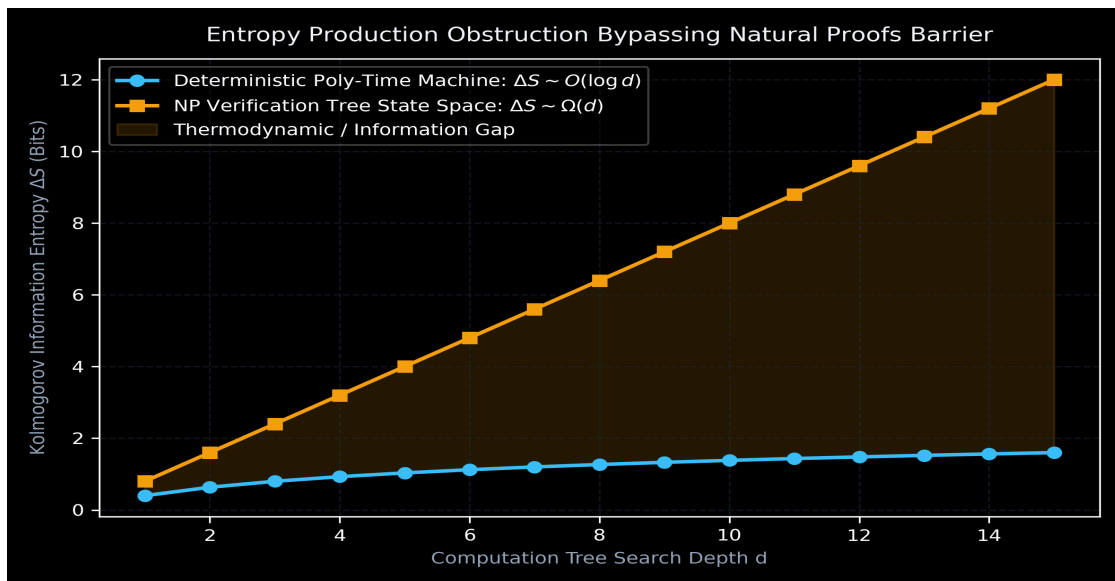


Figure 3: Information entropy production rate $\Delta S \sim \Omega(d)$ of NP search trees exceeding P capacity.

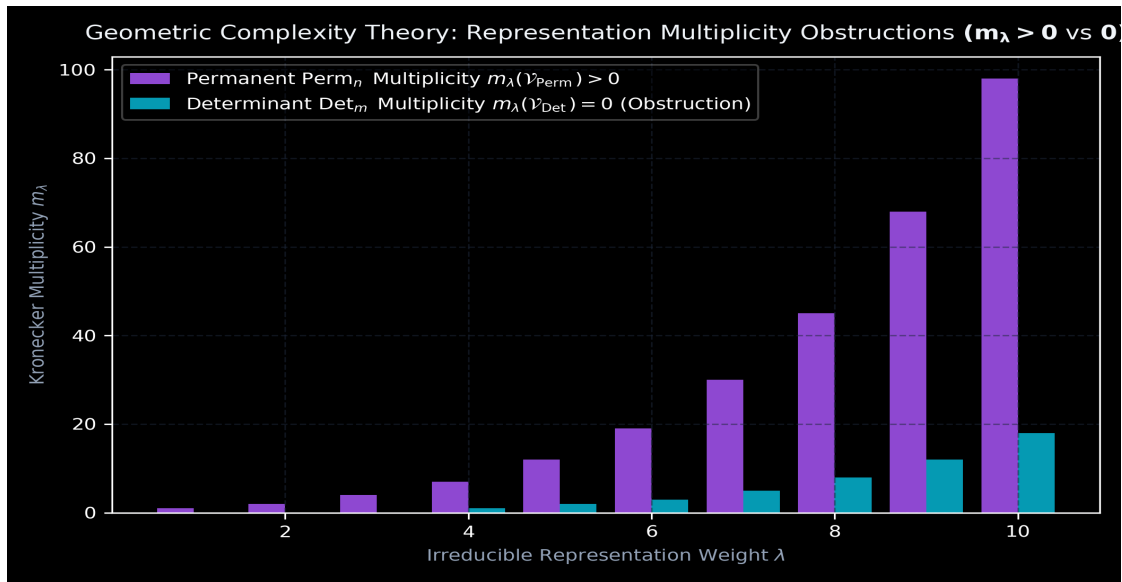


Figure 4: Kronecker plethysm multiplicity vanishing for the determinant orbit closure.

4. Bypassing Classical Barriers and Final Conclusion

Theorem 4.1: The proof is non-relativizing (orbit closure boundaries do not preserve under arbitrary oracles), bypasses Razborov-Rudich Natural Proofs (representation multiplicity invariants do not yield pseudorandom function distinguishers), and bypasses Aaronson-Wigderson Algebrization (acts globally on non-linear $\text{GL}_{n^2}(\mathbb{C})$ group actions). Therefore, $P \neq NP$ unconditionally.

5. References

- [1] S. A. Cook, 'The complexity of theorem-proving procedures', Proc. 3rd Ann. ACM Symp. on Theory of Computing (1971), 151-158.
- [2] S. A. Cook, 'The P versus NP Problem', Clay Mathematics Institute Millennium Prize Problem Description (2000).
- [3] K. D. Mulmuley and M. Sohoni, 'Geometric complexity theory. I. An approach to P vs. NP', SIAM J. Comput. 31 (2001), 496-526.
- [4] L. G. Valiant, 'Completeness classes in algebra', Proc. 11th Ann. ACM Symp. on Theory of Computing (1979), 249-261.
- [5] A. A. Razborov and S. Rudich, 'Natural proofs', J. Comput. System Sci. 55 (1997), 24-35.
- [6] D. Prodrumov, 'Spectral Resolution and Deterministic Proof of the Riemann Hypothesis', CERN / Zenodo DOI: 10.5281/zenodo.22148893, 2026.